

DBA-MA-CDB

Military Action Communications Design Basis

Sector-Agnostic Machinery Reference — Communications Degradation Baseline

Formalized from DBA-ES-MA-SF v0.2 and the superseded CB-1–CB-5 enumeration

Authored cross-sector for migration to the DBA-MA parent (Project 6)

Version 0.3 — DRAFT

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Sector-agnostic machinery reference for the DBA-MA series. Authored in Project 8 for migration to the DBA-MA parent. CDB-1–CDB-5 are a first formalization and remain subject to review; CDB-6 and CDB-7 restate obligations already in force.

1. Purpose and Scope

This reference module establishes the communications design basis for the Design Basis Accident — Military Action (DBA-MA) framework: the set of requirements that hold when a military-grade adversary degrades or denies the communications on which normal operations depend. It is authored to be sector-agnostic. The electric, natural gas, and petroleum sub-sector Military Action companions instantiate these requirements for their own facility classes rather than restating them.

DBA-ES-MA-SF v0.2 carries the governing principle — communications degradation is a design basis condition, not a contingency — distributed across its common attack surface analysis (§3.3), its control-coordination design basis assumptions (§5.4.3), and its design basis coherence review (§7.1). This module consolidates that distributed principle into an enumerated communications design basis. CDB-1 through CDB-5 formalize the distributed principle; CDB-6 and CDB-7 are carried forward from the superseded CB-1–CB-5 enumeration (Project 8 Instructions §2.3), preserving the Command Broker continuity and verification-cadence obligations that the distributed principle did not capture. The result is CDB-1 through CDB-7. Communications degradation is a denied condition (FD-BL-D1 §7); the continuity obligation and its mode of discharge are governed by FD-BL and FD-BL-D1, and the verification-cadence / indication-integrity requirement by FD-BL-D2, by reference.

1.1 The Governing Principle

The MA design basis treats communications degradation as a baseline condition, not a contingency. Several facility-class instruments developed earlier in the series implicitly assumed that normal communications would be available to support emergency procedures; instruments developed later treated communications degradation as a design basis condition that emergency procedures must accommodate. The MA design basis resolves that divergence in favor of the latter: degradation is assumed from the outset, and every requirement below follows from that assumption.

1.2 Naming Note: CDB versus CB

The communications design basis requirements are designated CDB-1 through CDB-7 to avoid collision with the Command Broker (CB) construct used in the DBA-MA parent series and its cross-sector companions. The two concepts intersect at two points — CDB-4 and CDB-6, both of which reference the Command Broker architecture. Keeping the communications requirements as CDB-* and the Command Broker as CB-* preserves the distinction precisely where the concepts touch. This designation supersedes the earlier “CB-1–CB-5” working vocabulary for the communications design basis; the two requirements unique to that earlier enumeration are preserved here as CDB-6 and CDB-7.

2. The Communications Design Basis (CDB-1 – CDB-7)

The following seven requirements constitute the communications design basis for the MA framework. Each applies to every facility class unless a sub-sector companion documents a justified exception. The source from which each requirement is formalized is noted for traceability.

CDB-1 — Degradation as baseline. All MA design basis analysis assumes adversary-imposed degradation or denial of normal communications as a starting condition, not a low-probability contingency. Emergency procedures, restoration plans, and coordination protocols are specified against the degraded condition and validated against it. (Source: §7.1, §3.3.)

CDB-2 — Local protective and safety autonomy. Loss of supervisory communications — the supervisory control and data acquisition (SCADA) and energy management system (EMS) visibility layer — must not cause loss of local protective, safety, and shutdown authority. Facility-level protective functions operate independently of the supervisory link, and the design basis must demonstrate that independence rather than assume it. (Source: §3.2, §5.1.3.)

CDB-3 — Field-executable emergency procedures. Emergency operating procedures must be executable by field operators acting without central or control-center direction. The design basis assumption that central direction will be available during an MA scenario is not valid and must not be relied upon in any emergency procedure specification. (Source: §5.4.3.)

CDB-4 — Verified independent channels. Independent communications channels between field facilities and system operators must be provided and verified as functional under degraded conditions. The Command Broker architecture is the reference framework for the human-confirmation channel this requirement depends upon; that channel must be local, auditable, and not itself executable by a software agent. (Source: §5.4.3.)

CDB-5 — Coordination without shared connectivity. Cross-operator and cross-sector coordination protocols must be executable without reliance on shared supervisory connectivity. Interface coordination — between operators, between sub-sectors, and across the gas-electric boundary in particular — must not assume that the shared SCADA or data layer remains available. (Source: §5.3.3, §6.6.)

CDB-6 — Command Broker continuity. Communications degradation does not suspend the Command Broker obligation. For any action above the Bright Line, the requirement that the Command Broker evaluate and authorize a Gen AI-recommended action before physical execution remains in force under degraded or denied communications. Degradation may not be invoked as grounds to bypass the Command Broker; the human-confirmation channel required by CDB-4 is the mechanism by which the obligation is met when supervisory communications are lost. (Source: former CB-3; cf. DI-5.)

CDB-7 — Verification cadence. The independent channels required by CDB-4 must be tested at defined intervals under conditions that simulate the MA communications baseline, with

primary supervisory communications assumed degraded or denied. Verification is a recurring obligation, not a one-time commissioning check. (Source: former CB-4.)

3. Instantiation Requirement

Each facility-class instrument applies CDB-1 through CDB-7 as design basis requirements specific to its facility class, identifying the local protective functions, field-executable procedures, independent channels, Command Broker interfaces, verification regime, and coordination interfaces that each requirement governs for that class. Where a facility class warrants a justified exception to any requirement, the exception is documented in the sub-sector Military Action companion, not taken silently at the instrument level. Earlier instruments that assumed communications availability in their emergency procedures are realigned to this baseline on their next revision, consistent with OI-1 of DBA-ES-MA-SF.

4. Provenance and Migration

CDB-1 through CDB-5 are a first formalization derived from DBA-ES-MA-SF v0.2 §§3.3, 5.4.3, and 7.1, supported by the design basis assumptions in §§5.1.3 and 5.3.3 and the interface-coordination requirement in §6.6. CDB-6 and CDB-7 are carried forward from the superseded CB-1–CB-5 enumeration (former CB-3, Command Broker continuity; former CB-4, verification cadence). Because the CDB-1–CDB-5 enumeration was newly constructed, it remains subject to review; CDB-6 and CDB-7 restate obligations already in force under the prior canon.

The module is authored for eventual migration to the DBA-MA parent series, where communications degradation is a cross-sector baseline. Before migration, the CDB-4 and CDB-6 references to the Command Broker architecture must be reconciled against the parent’s existing Command Broker framework (CB-Framework Cross-Sector v0.1) so that the CDB and CB constructs remain cleanly distinguished at the parent level.

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial version. First formalization of the MA communications design basis (CDB-1–CDB-5) from the distributed degradation-as-baseline principle in DBA-ES-MA-SF v0.2 §§3.3, 5.4.3, 7.1 (supported by §§5.1.3, 5.3.3, 6.6). Adopted the CDB-* prefix to resolve the collision with the Command Broker (CB) construct.
0.2	June 2026	T. Roxey	Absorbed former CB-3 (Command Broker continuity) and CB-4 (verification cadence) from the superseded CB-1–CB-5 enumeration as CDB-6 and CDB-7. Updated counts CDB-1–CDB-5 → CDB-1–CDB-7 throughout. Added the CDB-6 Command Broker intersection to the migration reconciliation note.

0.3	September 2026	T. Roxey	Cover disposition line revised for public issue; the requirements are unchanged. Issued to the publications mirror so that the Designation Register's supersession notes for CB-1–CB-5, which direct the reader to this module, resolve to an obtainable document. Version DOI to be minted in the next Zenodo batch.
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